

ACJC 2021 H2 Chemistry Prelim Practical

Prep List – Chemicals

Chemical Name	Qty per student	Mass / Concentration / Preparation method	Label for students
Part 1 – Volumetric Analysis			
H ₂ C ₂ O ₄ / Na ₂ C ₂ O ₄	220 cm ³	3.44 g dm ⁻³ H ₂ C ₂ O ₄ and 2.39 g dm ⁻³ Na ₂ C ₂ O ₄ (unknown to students)	FA 1
NaOH	100 cm ³	0.100 mol dm ⁻³	FA 2
KMnO ₄	150 cm ³	0.0200 mol dm ⁻³	FA 3
H ₂ SO ₄	150 cm ³	1 mol dm ⁻³	
MnSO ₄	50 cm ³	0.01 mol dm ⁻³	
Thymol blue			Shared on bench
Part 2 – Energetics and Planning			
H ₂ SO ₄	150 cm ³	1.00 mol dm ⁻³ (unknown to students)	FA 4
NaOH	180 cm ³	1.50 mol dm ⁻³	FA 5
Part 3 – Inorganic QA			
NH ₄ VO ₃ / ZnSO ₄ / H ⁺	10 cm ³	0.4 g NH ₄ VO ₃ + 3.0 g ZnSO ₄ to 100 mL + 10 mL 1.0M H ₂ SO ₄	FA 6
Barium chloride		0.5 mol dm ⁻³	Shared on bench
Magnesium turnings			Shared on bench
QA bench reagents			
NaOH		2 mol dm ⁻³	
NH ₃ (aq)		2 mol dm ⁻³	
Red litmus paper	2 pieces		Pack for students
Blue litmus paper	1 piece		
Wooden splint	2 pieces		

ACJC 2021 H2 Chemistry Prelim Practical

Prep List – Apparatus

Part 1 – Volumetric analysis

Per student	
1 x pipette filler	1 x retort stand with clamp
1 x 25 cm ³ pipette	1 x white tile
2 x 50 cm ³ burette with tap (no label)	2 x filter funnel
2 x 250 cm ³ conical flask	1 x wash bottle with deionised water
1 x 50 cm ³ measuring cylinder	1 x tripod
1 x 25 cm ³ measuring cylinder	1 x wire gauze
1 x 10 cm ³ measuring cylinder	1 x Bunsen burner
1 x thermometer	1 x gas lighter
1 x paper towel (fold to hold conical flask)	1 x goggles
1 x glass rod	1 x teat pipette (dropper) – replace end of shift

Part 2 – Energetics and Planning

Per student	
1 x 50 cm ³ burette with tap (no label)	Reuse from part 1
1 x filter funnel	
1 x 50 cm ³ measuring cylinder	
1 x 25 cm ³ measuring cylinder	
1 x 10 cm ³ measuring cylinder	
1 x thermometer	
1 x retort stand with clamp	
2 x teat pipettes (dropper)	Replace at end of each shift
2 x polystyrene cups	
1 x 250 cm ³ beaker	
1 x toilet roll (for drying)	

Part 3 – Inorganic QA

Per student	
1 x Bunsen burner	Reuse from part 1
1 x gas lighter	
1 x goggles	
1 x test-tube rack	
1 x test-tube holder	
1 x test-tube brush	
1 x wash bottle	Students refill at end of shift
1 x spatula	Replace at end of each shift
2 x wooden splints	
2 x red and 1 x blue litmus papers	
2 x test-tubes	
1 x boiling tube	
1 x teat pipette (dropper)	